

JACKSON, MN, USA

WMO: 726593

Lat: 43.650N

Lon: 94.986W

Elev: 441

StdP: 96.14

Time Zone: -6.00 (NAC)

Period: 98-19

WBAN: 04946

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.8	-21.1	-27.2	0.3	-22.4	-24.9	0.4	-20.9	15.4	-9.9	14.2	-9.5	4.3	310	0.616

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	32.3	22.9	31.0	22.1	28.7	21.0	25.3	29.5	24.0	28.0	22.9	26.9	5.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	20.0	27.2	22.6	18.3	26.5	22.0	17.6	26.0	80.3	29.3	74.6	28.0	70.1	26.6	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.7	11.3	10.1	-26.9	35.0	3.0	1.9	-29.1	36.4	-30.9	37.5	-32.6	38.5	-34.8	39.9		
(5)			-27.3	27.1	3.2	1.9	-29.6	28.5	-31.5	29.6	-33.3	30.6	-35.6	32.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.9	-8.3	-6.4	0.5	7.9	14.5	20.1	22.2	20.4	17.0	9.3	2.0	-5.8
	DBStd	12.08	7.12	6.90	6.87	5.64	5.11	3.74	3.29	3.20	4.75	5.42	5.95	6.64
	HDD10.0	2270	567	459	303	103	17	0	0	0	5	78	247	491
	HDD18.3	4171	825	692	554	314	140	24	6	14	80	283	490	748
	CDD10.0	1491	0	0	8	41	157	304	378	324	214	58	7	0
	CDD18.3	351	0	0	0	2	22	78	126	80	40	4	0	0
	CDH23.3	3184	0	0	3	32	249	702	1131	646	369	51	1	0
	CDH26.7	941	0	0	0	7	76	225	358	166	97	12	0	0
Wind	WSAvg	4.7	5.4	5.3	5.3	5.5	5.2	4.3	3.4	3.1	3.8	4.6	5.1	5.1
Precipitation	PrecAvg	769	18	18	42	82	105	121	100	94	75	56	34	23
	PrecMax	1151	37	45	79	165	173	260	195	250	222	166	103	61
	PrecMin	516	2	1	2	25	33	59	17	18	9	8	2	3
	PrecStd	155	10	11	22	36	41	48	45	52	58	40	26	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.9	12.5	22.4	27.4	32.2	33.0	33.9	32.8	32.3	27.8	21.0	12.2
		MCWB	3.9	8.2	13.4	15.4	19.1	21.7	24.6	25.2	20.8	18.6	12.2	9.7
	2%	DB	4.0	7.8	17.3	22.9	28.6	31.3	32.3	30.2	29.8	24.0	16.3	7.7
		MCWB	1.8	5.5	11.5	12.8	17.4	20.6	24.3	23.6	20.8	16.2	10.7	4.6
	5%	DB	2.4	4.1	12.8	20.8	26.2	29.1	30.9	27.9	27.3	21.2	12.9	4.1
		MCWB	1.2	2.5	9.4	12.2	16.4	19.6	23.6	22.0	19.0	14.1	8.0	1.8
	10%	DB	1.0	2.4	10.1	17.4	22.9	27.3	28.0	27.1	25.8	17.8	10.8	2.4
		MCWB	0.3	1.3	7.4	10.5	14.8	19.1	21.8	21.3	18.2	12.4	7.2	1.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.4	9.7	16.5	16.9	21.8	24.1	27.5	27.6	24.0	22.5	13.9	10.1
		MCDB	7.6	12.8	20.4	23.4	27.9	29.5	31.1	31.0	28.8	25.5	18.5	12.1
	2%	WB	2.5	5.6	12.6	14.7	19.7	22.9	25.9	25.1	22.5	17.4	11.3	4.6
		MCDB	3.9	8.0	15.3	20.5	25.3	28.3	29.9	28.3	27.1	20.2	15.5	7.3
	5%	WB	1.2	2.9	10.1	13.2	18.0	21.7	24.5	23.6	21.1	15.1	8.9	2.6
		MCDB	2.3	4.3	13.5	18.8	23.3	26.6	28.8	26.8	25.1	19.4	12.7	4.1
	10%	WB	0.1	1.3	7.1	11.6	16.5	20.5	23.4	22.3	19.7	13.0	6.9	1.1
		MCDB	0.8	2.2	9.9	16.0	21.1	25.3	27.4	25.8	23.8	17.6	10.3	2.2
Mean Daily Temperature Range	5% DB	MDBR	8.4	8.6	8.8	11.0	11.5	10.9	10.8	10.7	12.2	11.0	9.3	7.9
		MCDBR	9.5	10.3	14.0	16.4	15.2	13.5	11.7	11.8	14.0	15.2	13.4	10.5
		MCWBR	7.8	7.9	8.8	8.5	6.9	5.7	6.1	6.5	6.7	8.0	8.1	8.0
	5% WB	MCDBR	8.6	9.8	13.0	13.5	12.4	10.7	10.6	10.4	11.9	12.2	12.6	9.8
		MCWBR	7.3	7.7	8.7	8.2	6.7	6.0	6.4	6.6	6.9	7.7	8.2	7.7
Clear-Sky Solar Irradiance	taub	0.267	0.281	0.310	0.359	0.379	0.398	0.410	0.405	0.362	0.323	0.284	0.265	
	taud	2.440	2.419	2.443	2.339	2.350	2.353	2.302	2.295	2.389	2.489	2.522	2.471	
	Ebn at Noon	878	923	934	905	892	874	857	849	863	861	852	847	
	Edh at Noon	72	88	99	119	121	122	127	123	104	82	67	64	
All-Sky Solar Radiation	RadAvg	1.86	2.93	3.83	4.71	5.50	6.10	6.54	5.42	4.34	2.89	1.94	1.43	
	RadStd	0.20	0.31	0.36	0.44	0.39	0.45	0.26	0.40	0.29	0.37	0.15	0.14	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (58 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page